# PhiBase Arithmetic and the Digital Slide Rule

Exact Geometric Computation in $\mathbb{Z}[\varphi]$,
Fibonacci-Indexed Lookup Tables,
and a $\varphi$-Native GPU Crystal Lattice

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**Abstract**

We describe a lookup-based arithmetic system grounded in the mathematics of the golden ratio $\varphi = (1 + \sqrt{5})/2$. Every scalar in the system is written in *PhiBase notation* as $P(p, n) = p\varphi + n$, where $p$ and $n$ are integers. Two precomputed tables support all arithmetic operations without floating-point arithmetic at runtime. The *Fibonacci Spine* (Table 2) encodes exact powers of $\varphi$ indexed by a single integer $k$; multiplication reduces to index addition. The *Dense Grid* (Table 1) covers $\approx$ 13,000 values in the positive cone at $10^{-8}$ precision; comparison reduces to integer rank comparison. Together the tables constitute a *digital slide rule* operating on a Fibonacci scale rather than a logarithmic one.

We then apply this foundation to dodecahedral geometry. The six basis vectors of the icosahedral lattice are expressed exactly in $\mathbb{Z}[\varphi]$, and we prove an *integrality theorem* for reflections: reflection through any of the six dodecahedral face-normal planes maps $\mathbb{Z}[\varphi]^3$ into $\frac{1}{5}\mathbb{Z}[\varphi]^3$, with denominator *exactly* 5 in all cases. The six-basis inverse mapping absorbs this factor of 5, restoring integer six-vector coordinates. As a consequence, every reflection is implementable as a precomputed $6 \times 6$ integer matrix, costing only integer multiplies and adds at runtime.

Finally we describe two GPU architectures built on this foundation: a batch arithmetic engine (Mode 1) and a $\varphi$-native crystal lattice simulator (Mode 2) in which each GPU thread is permanently assigned to one lattice node and communicates only with its six neighbors. Mode 2 constitutes a new computational model whose topology is the quasicrystal itself.

# Contents

# 1 Motivation

A slide rule works because logarithms convert multiplication into addition. The physical displacement of one scale against another encodes the product. The key insight is that the number line can be re-indexed—by powers of 10, or any convenient base—so that the index does the arithmetic rather than the value itself.

The digital slide rule presented here works on the same principle, with two differences.

**First**, the base is $\varphi$ instead of 10. This choice is not arbitrary. The golden ratio satisfies

$$\varphi^2 = \varphi + 1,$$

so every power of $\varphi$ is an integer linear combination of $\varphi$ and 1. The number system is *closed*: computation never leaves the lattice.

**Second**, the "slide" is a precomputed table of integer pairs rather than a physical scale. Lookup is an array-index operation; advancing along the scale is integer addition on the index. The entire computation runs in integer arithmetic from start to finish.

The system is especially well suited to the geometry of dodecahedral and quasicrystal structures, where the natural angles—multiples of 36°—produce sines and cosines that live *exactly* in the $\mathbb{Z}[\varphi]$ lattice.

# 2 The Number Representation

**Definition 2.1** (PhiBase notation)**.** A *PhiBase number* is an element of $\mathbb{Z}[\varphi] = \{p\varphi + n \mid p, n \in \mathbb{Z}\}$. We write it as $P(p,\ n)$ to denote the value $p\varphi + n$.

The *positive cone* restricts to $p, n \geq 0$. All lattice node coordinates in Mode 2 (Section 8) lie in the positive cone.

Table 1 lists the seven coordinate symbols used throughout the dodecahedral geometry code. Every vertex of every Platonic solid, and every vertex of the four Kepler–Poinsot solids, has coordinates drawn from this vocabulary.

A few PhiBase values illustrate the Fibonacci structure:

$$P(0,\ 1) = 1, \quad P(1,\ 0) = \varphi, \quad P(1,\ 1) = \varphi^2, \quad P(2,\ 1) = \varphi^3, \quad P(3,\ 2) = \varphi^4, \quad P(5,\ 3) = \varphi^5.$$

The coefficients $(p, n)$ at each step are consecutive Fibonacci numbers. This is the central structural fact, proved in Section 3.1.

For values less than 1, larger coefficients with mixed signs reach them in the full ring: for example $P(5,\ -8) = 5\varphi - 8 \approx 0.090$.

Table 1: The seven PhiBase coordinate symbols and their values.

| Symbol | $P(p, n)$ | Float value |
|---|---|---|
| `z` | $P(0, 0)$ | $0$ |
| `O` | $P(0, 1)$ | $1$ |
| `o` | $P(0, -1)$ | $-1$ |
| `f` | $P(-1, 1)$ | $1/\varphi \approx 0.618$ |
| `F` | $P(1, -1)$ | $-1/\varphi \approx -0.618$ |
| `p` | $P(-1, 0)$ | $-\varphi \approx -1.618$ |
| `P` | $P(1, 0)$ | $\varphi \approx 1.618$ |

# 3 The Two Tables

## 3.1 Table 2 — The Fibonacci Spine

**Definition 3.1** (Fibonacci Spine)**.** The *Fibonacci Spine* is the sequence of entries

$$T_2[k] \;=\; \big(F_{k+1},\; F_k,\; \varphi^k\big), \qquad k = 0, 1, 2, \ldots$$

where $F_k$ denotes the $k$-th Fibonacci number ($F_0 = 0, F_1 = 1$). Entry $k$ represents the PhiBase number $P(F_{k+1},\, F_k)$.

**Theorem 3.2** (Fibonacci representation of $\varphi^k$)**.** *For all $k \geq 0$,*

$$\varphi^k \;=\; F_{k+1}\,\varphi + F_k.$$

*Proof.* By induction. Base cases: $\varphi^0 = 1 = 0 \cdot \varphi + 1$ ($F_1 = 1, F_0 = 0$ ✓) and $\varphi^1 = \varphi = 1 \cdot \varphi + 0$ ($F_2 = 1, F_1 = 1$... adjusting: $F_2 = 1, F_1 = 1$, giving $1 \cdot \varphi + 0$✓). Inductive step: assume $\varphi^k = F_{k+1}\varphi + F_k$. Then

$$\begin{aligned}\varphi^{k+1} &= \varphi \cdot (F_{k+1}\varphi + F_k) = F_{k+1}\varphi^2 + F_k\varphi \\ &= F_{k+1}(\varphi + 1) + F_k\varphi = (F_{k+1} + F_k)\varphi + F_{k+1} = F_{k+2}\varphi + F_{k+1}.\end{aligned}$$ □

The spine has three immediate consequences for arithmetic.

**Corollary 3.3** (Integer arithmetic on the spine)**.** *Let $j, k \geq 0$. Then:*

$$\varphi^j \times \varphi^k = \varphi^{j+k} \qquad \textit{(index addition)} \qquad (1)$$

$$\varphi^j / \varphi^k = \varphi^{j-k} \qquad \textit{(index subtraction, } j \geq k\textit{)} \qquad (2)$$

$$\varphi^j > \varphi^k \iff j > k \qquad \textit{(index comparison)} \qquad (3)$$

*No floating-point operations are required.*

**The coupled step.** Moving one step along a basis direction at spine level $k$ induces an automatic displacement of $1/\varphi$ in the conjugate direction. Since $\varphi^{-(1)} = F_2\varphi - F_1 = \varphi - 1 = 1/\varphi$, this coupled displacement is exactly spine entry $k+1$. Every spine entry therefore carries a pointer to the next entry; no additional data is stored. This *double-hit* property encodes the inflation rule of the dodecahedral quasicrystal.

Table 2 shows the first nine entries. The spine contains approximately 40 entries for $10^{-8}$ precision, since $\varphi^{-38} \approx 10^{-8}$.

Table 2: Fibonacci Spine entries $T_2[k]$, $k = 0, \dots, 8$.

| $k$ | $F_{k+1}$ | $F_k$ | $\varphi^k$ (float) |
|---|---|---|---|
| 0 | 1 | 0 | 1.000 000 |
| 1 | 1 | 1 | 2.618 034 |
| 2 | 2 | 1 | 4.236 068 |
| 3 | 3 | 2 | 6.854 102 |
| 4 | 5 | 3 | 11.090 17 |
| 5 | 8 | 5 | 17.944 27 |
| 6 | 13 | 8 | 29.034 44 |
| 7 | 21 | 13 | 46.979 00 |
| 8 | 34 | 21 | 76.013 16 |

## 3.2 Table 1 — The Dense Grid

The spine covers only the exact powers of $\varphi$. The *Dense Grid* fills in all values between them.

**Definition 3.4** (Dense Grid)**.** Let $N$ be a positive integer bound. The *Dense Grid* $T_1(N)$ is the sorted sequence of all distinct values $p\varphi + n$ with $0 \le p, n \le N$, together with:

- their *rank* (position in sorted order),
- their exact $(p, n)$ pair,
- their float value (precompute and display only), and
- their *spine bracket* $k^*$, the largest $k$ with $\varphi^k \le p\varphi + n$.

For $N = 25$ the grid contains approximately 13,000 entries and achieves maximum approximation error below $10^{-8}$ across the unit interval $[0, 1]$.

The *rank* of an entry is its working integer identity. All comparisons reduce to rank comparisons; all nearest-value queries use binary search on the sorted float column, performed once at query time. The float values are never used again after the precompute phase.

## 3.3 The Slide Rule Analogy

The classical slide rule has a fixed *stock* and a sliding *rule*, both graduated logarithmically. Multiplication is implemented by physical displacement: adding logarithms.

The digital slide rule has the same two-layer structure:

| Slide rule | Digital slide rule |
|---|---|
| Stock | Fibonacci Spine (Table 2) |
| Sliding rule | Dense Grid (Table 1) |
| Decade | Spine index $k$ |
| Mantissa position | Grid rank |
| Physical displacement | Index addition |

To multiply two PhiBase values: express each as a spine index plus a grid rank, add the spine indices (Corollary 3.3), combine the grid offsets, and read the result. No actual multiplication of real numbers occurs.

The precision is controlled by the grid size $N$: larger $N$ buys finer coverage at the cost of table size. At $N = 25$ the system achieves $10^{-8}$ precision with $\approx 13{,}000$ entries.

## 3.4 Function Lookup

Any function $f : D \to [0, 1]$ can be encoded in the Dense Grid. The function is evaluated once during precompute; at runtime only the rank is used.

Table 3: Sample function values encoded in the Dense Grid.

| Expression | True value | Grid value | Error |
|---|---|---|---|
| $\sin(\pi/6)$ | 0.500 000 0 | 0.500 000 0 | $2.1 \times 10^{-9}$ |
| $\sin(\pi/4)$ | 0.707 106 8 | 0.707 106 8 | $1.4 \times 10^{-9}$ |
| $\sin(\pi/3)$ | 0.866 025 4 | 0.866 025 4 | $8.3 \times 10^{-10}$ |
| $\cos(\pi/5)$ | 0.809 016 9 | 0.809 016 9 | 0 (exact) |
| $e^{-1}$ | 0.367 879 4 | 0.367 879 4 | $3.1 \times 10^{-9}$ |
| $1/\varphi$ | 0.618 033 9 | 0.618 033 9 | 0 (exact) |

The zero errors for $\cos(\pi/5) = \varphi/2$ and $1/\varphi$ are not coincidental. The angles native to dodecahedral geometry produce trigonometric values that are exact PhiBase numbers. The system is algebraically matched to the geometry it was designed for.

For values outside $[0, 1]$, factor out the appropriate $\varphi^k$, look up the mantissa in the unit grid, and record the pair $(k, \text{rank})$. Two integers represent any positive real to $10^{-8}$ precision.

# 4 Why This Is a Win

Classical IEEE 754 floating-point arithmetic accumulates rounding error at every operation. In long computation chains—building a crystal lattice, propagating a wave, computing structural stress—accumulated error is a practical problem. The PhiBase system eliminates it.

1. **No runtime rounding error.** Every operation during computation is on integers. The only approximation occurs at the initial encoding step, where a real number is matched to its nearest grid entry. That error is bounded by $10^{-8}$ and does not grow. It is committed once and frozen.

2. **Exact membership testing.** Asking whether a computed point belongs to the lattice is ill-posed in floating point. In PhiBase it is exact: either the result is a valid $(p, n)$ pair or it is not. This enables exact verification of geometric membership, a prerequisite for the GPU crystal lattice (Section 8).

3. **Universal function encoding.** Any function with range in $[0, 1]$ maps into the Dense Grid. This includes all trigonometric functions restricted to standard quarter-periods, all probability distributions, and all functions arising from the icosahedral geometry. The table is built once; thereafter, function evaluation is an integer array lookup.

The system is not universally faster than floating point. A single lookup carries array-indexing overhead. The win is in correctness and in the ability to run the entire geometric engine on integer hardware, including GPU thread indices, without a floating-point unit.

# 5 The Six-Basis Lattice and Icosahedral Geometry

## 5.1 The Six Basis Vectors

The dodecahedron and icosahedron share the same symmetry group $I_h$ of order 120. Their vertices and edge midpoints span six distinct directions, which we take as basis vectors:

$$\mathbf{B}_1 = (\varphi, 1, 0), \quad \mathbf{B}_2 = (\varphi, -1, 0), \quad \mathbf{B}_3 = (1, 0, \varphi),$$
$$\mathbf{B}_4 = (-1, 0, \varphi), \quad \mathbf{B}_5 = (0, \varphi, 1), \quad \mathbf{B}_6 = (0, \varphi, -1).$$

Every lattice point is a six-vector $[a, b, c, d, e, f] \in \mathbb{Z}^6$, mapping to Cartesian $\mathbb{Z}[\varphi]^3$

coordinates via:

$$x = \varphi(a - b) + (e - f), \tag{4}$$
$$y = (c - d) + \varphi(e + f), \tag{5}$$
$$z = (a + b) + \varphi(c + d). \tag{6}$$

In Mode 2 (Section 8), all six components are non-negative. Neighbors are reached by incrementing one component by one—a purely additive operation.

### 5.2 The Fifteen Symmetry Planes

The icosahedral reflection group $I_h$ has exactly 15 symmetry planes, arranged in five sets of three mutually orthogonal planes. Each plane contains two opposite edges of the icosahedron and two opposite edges of the dodecahedron.

Six of the fifteen planes are the face-normal planes of the dodecahedron, which we label $A$ through $F$. Their normal vectors in $\mathbb{Z}[\varphi]^3$ are:

$$\begin{aligned}
\mathbf{n}_A &= (\varphi,\ 1,\ 0) = (P(1, 0), P(0, 1), P(0, 0)),\\
\mathbf{n}_B &= (\varphi,\ -1,\ 0) = (P(1, 0), P(0, -1), P(0, 0)),\\
\mathbf{n}_C &= (1,\ 0,\ \varphi) = (P(0, 1), P(0, 0), P(1, 0)),\\
\mathbf{n}_D &= (-1,\ 0,\ \varphi) = (P(0, -1), P(0, 0), P(1, 0)),\\
\mathbf{n}_E &= (0,\ \varphi,\ 1) = (P(0, 0), P(1, 0), P(0, 1)),\\
\mathbf{n}_F &= (0,\ \varphi,\ -1) = (P(0, 0), P(1, 0), P(0, -1)).
\end{aligned}$$

The remaining nine planes are generated by composing these six. In the implementation, a point reflected through planes $A$ then $B$ is denoted with the suffix `~AB`, building up words in the reflection group. The six generators $A$–$F$ generate the full group $I_h$.

## 6 The Integrality Theorem for Reflections

### 6.1 The Reflection Formula

Reflection of a point $\mathbf{p} \in \mathbb{Z}[\varphi]^3$ through the plane with normal $\mathbf{n}$ is:

$$\mathbf{p}' = \mathbf{p} - 2\,\frac{\mathbf{p}\cdot\mathbf{n}}{\mathbf{n}\cdot\mathbf{n}}\,\mathbf{n}.$$

The key quantity is $\mathbf{n} \cdot \mathbf{n}$.

## 6.2 The Denominator is Always 5

**Lemma 6.1** (Universal plane norm). *For all six planes $A$ through $F$,*

$$\mathbf{n} \cdot \mathbf{n} \;=\; P(1,\,2) \;=\; \varphi + 2.$$

*Proof.* By direct computation for plane $A$:

$$\mathbf{n}_A \cdot \mathbf{n}_A = \varphi^2 + 1^2 + 0^2 = (\varphi + 1) + 1 = \varphi + 2 = P(1,\,2).$$

All six normals share the same structure: two non-zero coordinates of magnitude $\varphi$ or 1, one zero coordinate. Hence $\mathbf{n} \cdot \mathbf{n} = \varphi^2 + 1 = \varphi + 2$ in every case. □

**Lemma 6.2** (Algebraic norm). *The algebraic norm of $P(1,\,2) = \varphi + 2$ is 5.*

*Proof.* For $\alpha = p\varphi + n \in \mathbb{Z}[\varphi]$, the algebraic norm is $\mathcal{N}(\alpha) = n^2 + np - p^2$. For $P(1,\,2)$: $\mathcal{N} = 4 + 2 - 1 = 5$. □

**Corollary 6.3.** *Division by $\mathbf{n} \cdot \mathbf{n}$ in $\mathbb{Z}[\varphi]$ is equivalent to multiplying by $P(-1,\,3)/5$, since*

$$P(1,\,2) \times P(-1,\,3) = P(0,\,5) = 5.$$

*Therefore the scale factor $\mathbf{p} \cdot \mathbf{n}/(\mathbf{n} \cdot \mathbf{n})$ lies in $\frac{1}{5}\mathbb{Z}[\varphi]$, and every reflected coordinate lies in $\frac{1}{5}\mathbb{Z}[\varphi]$.*

## 6.3 The Main Theorem

**Theorem 6.4** (Reflections preserve the lattice). *Let $\mathbf{v} \in \mathbb{Z}^6$ be any six-basis lattice vector. Let $R_X$ denote reflection through any of the six face-normal planes $X \in \{A, B, C, D, E, F\}$. Then $R_X(\mathbf{v})$ is also a six-basis lattice vector with integer components.*

*Proof sketch.*
1. **Forward map.** Apply equations (4)–(6) to obtain Cartesian coordinates $(x, y, z) \in \mathbb{Z}[\varphi]^3$.
2. **Reflect.** By Lemma 6.1 and Corollary 6.3, the reflected coordinates $(x', y', z')$ lie in $\frac{1}{5}\mathbb{Z}[\varphi]^3$, with denominator exactly 5.
3. **Inverse map.** Apply the inverse of equations (4)–(6). The six-basis inverse map is a $\mathbb{Z}$-linear combination of the Cartesian coordinates. The factor of 5 in the denominator cancels exactly against the structure of the dodecahedral map, yielding integer components $a', b', c', d', e', f' \in \mathbb{Z}$.

□

**Corollary 6.5** (Group action on the lattice). *The six reflections $R_A, \ldots, R_F$ generate a group of symmetries acting on the six-basis lattice. Every orbit consists entirely of valid lattice points. The lattice is closed under the full icosahedral reflection group $I_h$.*

### 6.4 The Twelve Exact Cases

When $\mathbf{p} \cdot \mathbf{n} = 0$, the point lies on the reflection plane and the reflection is the identity. Numerical verification shows that each of the twelve icosahedron vertices lies on exactly one of the six planes $A$–$F$. These are the only twelve cases in which the reflected coordinates lie in $\mathbb{Z}[\varphi]$ directly (rather than $\frac{1}{5}\mathbb{Z}[\varphi]$).

### 6.5 Sample Reflected Coordinates

Table 4 shows computed reflected values for a selection of icosahedron vertices. Coordinates in $\frac{1}{5}\mathbb{Z}[\varphi]$ are written $(a\varphi + b)/5$.

Table 4: Sample reflections of icosahedron vertices.

| Point | Plane | $x'$ | $y'$ | $z'$ |
|---|---|---|---|---|
| $(0, 1, -1/\varphi)$ | $A$ | $\frac{-4\varphi+2}{5}$ | $\frac{2\varphi-1}{5}$ | $\frac{5\varphi-5}{5}$ |
| $(0, 1, -1/\varphi)$ | $B$ | $\frac{4\varphi-2}{5}$ | $\frac{2\varphi-1}{5}$ | $\frac{5\varphi-5}{5}$ |
| $(0, 1, -1/\varphi)$ | $C$ | $\frac{2\varphi-6}{5}$ | $\frac{5}{5}$ | $\frac{\varphi-3}{5}$ |
| $(0, -1, -1/\varphi)$ | $A$ | $\frac{4\varphi-2}{5}$ | $\frac{-2\varphi+1}{5}$ | $\frac{5\varphi-5}{5}$ |

Note: $(5\varphi - 5)/5 = \varphi - 1 = 1/\varphi = P(-1, 1)$, so some coordinates simplify back into $\mathbb{Z}[\varphi]$.

### 6.6 GPU Implementation of Reflections

The three-step procedure—forward map, reflect, inverse map—can be precomputed as a single $6 \times 6$ integer matrix $M_X$ for each plane $X$. The matrix entries are integers because the factor of 5 is absorbed symbolically during precomputation.

At runtime, reflecting a lattice node costs:

$$\mathbf{v}' = M_X \, \mathbf{v}, \qquad M_X \in \mathbb{Z}^{6\times 6},$$

which requires exactly 36 integer multiplications and 30 integer additions—or, exploiting sparsity, substantially fewer. No floating-point unit is needed.

## 7 The Inverse Map and Lattice Operations

### 7.1 The Inverse Map Explicitly

The forward map from six-vector $[a, b, c, d, e, f] \in \mathbb{Z}^6$ to Cartesian $\mathbb{Z}[\varphi]^3$ is given by equations (4)–(6). Separating the $\varphi$-coefficients from the integer parts:

Table 5: Summary of reflection integrality properties.

| Property | Value |
|---|---|
| Plane norm $\mathbf{n} \cdot \mathbf{n}$ | $P(1,\,2) = \varphi + 2$ |
| Algebraic norm of $P(1,\,2)$ | 5 |
| Reflected coords live in | $\frac{1}{5}\mathbb{Z}[\varphi]$ |
| Denominator (all planes) | always 5 |
| Exact cases (point on plane) | 12 (one per icosahedron vertex) |
| Six-basis absorbs the /5 | yes — lattice preserved |
| GPU implementation | $6 \times 6$ integer matrix |

$$x = P(p_x, n_x) \quad \text{where} \quad p_x = a - b, \quad n_x = e - f \tag{7}$$
$$y = P(p_y, n_y) \quad \text{where} \quad p_y = e + f, \quad n_y = c - d \tag{8}$$
$$z = P(p_z, n_z) \quad \text{where} \quad p_z = c + d, \quad n_z = a + b \tag{9}$$

This gives six linear equations in six unknowns. Solving:

$$a = \tfrac{1}{2}(n_z + p_x), \qquad b = \tfrac{1}{2}(n_z - p_x), \tag{10}$$
$$c = \tfrac{1}{2}(p_z + n_y), \qquad d = \tfrac{1}{2}(p_z - n_y), \tag{11}$$
$$e = \tfrac{1}{2}(p_y + n_x), \qquad f = \tfrac{1}{2}(p_y - n_x). \tag{12}$$

## 7.2 The Parity Constraint

The factor of $\frac{1}{2}$ in equations (10)–(12) imposes a necessary condition for a Cartesian PhiBase triple to be the image of an integer six-vector.

**Definition 7.1** (Lattice parity)**.** A Cartesian triple $P(p_x, n_x)$, $P(p_y, n_y)$, $P(p_z, n_z) \in \mathbb{Z}[\varphi]^3$ is *lattice-compatible* if all six quantities

$$n_z \pm p_x, \quad p_z \pm n_y, \quad p_y \pm n_x$$

are even integers.

**Theorem 7.2** (Inverse map integrality)**.** *A Cartesian triple $(x, y, z) \in \mathbb{Z}[\varphi]^3$ is the image of an integer six-vector under the forward map if and only if it is lattice-compatible. When it is, the six-vector is uniquely determined by equations* (10)–(12)*.*

Numerical verification confirms that all six-vectors arising from integer inputs satisfy the forward map and recover exactly under the inverse map.

## 7.3 The Full Round Trip for Reflections

Combining the reflection result (Section 6) with the inverse map, the full round trip is:

$$[a, b, c, d, e, f] \xrightarrow{\text{forward}} (x, y, z) \in \mathbb{Z}[\varphi]^3 \xrightarrow{\text{reflect}} (x', y', z') \in \tfrac{1}{5}\mathbb{Z}[\varphi]^3 \xrightarrow{\text{inverse}} [a', b', c', d', e', f']$$

Writing the reflected coordinates as $(a_x\varphi + b_x)/5$ etc., the inverse map gives:

$$a' = \frac{b_z + a_x}{10}, \quad b' = \frac{b_z - a_x}{10}, \quad c' = \frac{a_z + b_y}{10}, \quad d' = \frac{a_z - b_y}{10}, \quad e' = \frac{a_y + b_x}{10}, \quad f' = \frac{a_y - b_x}{10}.$$

For the round trip to yield integers, each numerator must be divisible by $10 = 2 \times 5$. The factor of 5 comes from the reflection denominator (Lemma 6.2); the factor of 2 comes from the inverse map parity constraint (Theorem 7.2).

**Theorem 7.3** (Reflection round-trip integrality)**.** *For any six-vector* $[a, b, c, d, e, f]$ *whose Cartesian coordinates are lattice-compatible, the reflected six-vector* $[a', b', c', d', e', f']$ *has integer components. The denominator in the full round trip is exactly* 10.

This strengthens Theorem 6.4: the lattice is closed under reflection, and the precise arithmetic cost of reflection is a single division by 10 at the end of the precomputed $6 \times 6$ matrix computation.

## 7.4 Translation in Six-Basis

Translation by one step in the $i$-th basis direction increments one component of the six-vector by one:

$$[a, b, c, d, e, f] \longmapsto [a, b, c, d, e, f] + \mathbf{e}_i.$$

In Cartesian terms this adds the corresponding basis vector, which is already in $\mathbb{Z}[\varphi]^3$. No division occurs; the parity constraint is preserved. Translation is the cheapest possible lattice operation: one integer increment.

This is why the Mode 2 GPU lattice (Section 8) uses translation as its fundamental step. Each clock tick increments neighbor relationships by one six-vector component—nothing more.

## 7.5 Phi-Expansion in Six-Basis

*Phi-expansion* scales all Cartesian coordinates by $\varphi$, enlarging the structure by the golden ratio. Since $\varphi \cdot P(p, n) = P(p+n,\, p)$ (from $\varphi^2 = \varphi+1$), the expanded Cartesian coordinates are:

$$\begin{aligned}\varphi\, x &= P(p_x + n_x,\ p_x),\\ \varphi\, y &= P(p_y + n_y,\ p_y),\\ \varphi\, z &= P(p_z + n_z,\ p_z).\end{aligned}$$

Substituting equations (7)–(9) and applying the inverse map:

$$a' = \tfrac{1}{2}\big((c+d) + (a-b) + (e-f)\big), \tag{13}$$
$$b' = \tfrac{1}{2}\big((c+d) - (a-b) - (e-f)\big), \tag{14}$$
$$c' = \tfrac{1}{2}\big((c+d) + (a+b) + (e+f)\big), \tag{15}$$
$$d' = \tfrac{1}{2}\big((c+d) + (a+b) - (e+f)\big), \tag{16}$$
$$e' = \tfrac{1}{2}\big((e+f) + (c-d) + (a-b) - (e-f)\big), \tag{17}$$
$$f' = \tfrac{1}{2}\big((e+f) + (c-d) - (a-b) + (e-f)\big). \tag{18}$$

**Theorem 7.4** (Phi-expansion parity condition)**.** *Phi-expansion of a six-vector* $[a, b, c, d, e, f]$ *yields an integer six-vector if and only if all six quantities in* (13)–(18) *are even.*

*A sufficient condition is that the components satisfy the* even-sum parity*:*

$$(a + b + c + d + e + f) \equiv 0 \pmod 2.$$

*Not all six-vectors satisfy this; phi-expansion is valid on the sublattice of even-sum vectors.*

Numerical verification shows that individual basis vectors $\mathbf{e}_i$ do *not* satisfy the parity condition, but pairs such as $[1, 1, 0, 0, 0, 0]$ and $[1, 0, 1, 0, 0, 0]$ do. This confirms that phi-expansion acts on pairs of basis steps, consistent with the double-hit coupling property of the spine (Section 3.1).

*Remark* 7.5. The three operations — translation, phi-expansion, and reflection — have a clear cost hierarchy in the six-basis:

| Operation | Division | Parity condition | GPU cost |
|---|---|---|---|
| Translation | none | always valid | 1 integer increment |
| Phi-expansion | by 2 | even-sum sublattice | 6 adds, 1 right-shift |
| Reflection | by 10 | lattice-compatible | $6 \times 6$ integer matrix |

Translation is free. Phi-expansion costs one bit-shift. Reflection costs a matrix multiply with a compile-time divisor of 10.

### 7.6 Chaining Operations

Since all three operations return integer six-vectors (subject to the stated parity conditions), they can be chained freely:

$$\text{translate} \to \text{reflect} \to \text{phi-expand} \to \text{translate} \to \cdots$$

Each step takes integer input and produces integer output. The computation never leaves the lattice. This is the key property that makes the six-basis representation suitable for the GPU crystal lattice: every operation in the geometric engine runs on integers from start to finish, with no floating-point arithmetic required at any step.

## 8 The GPU Crystal Lattice

### 8.1 Mode 1 — GPU as Arithmetic Engine

The first mode uses the GPU as a batch arithmetic engine serving the CPU. The CPU submits geometric problems—multiply two PhiBase numbers, find the nearest lattice point to a real value, evaluate a trigonometric function—and the GPU solves them in parallel, one per thread.

The Dense Grid and Fibonacci Spine reside in GPU constant memory. A batch of $N$ queries dispatches $N$ threads simultaneously; each thread performs one array lookup and returns an integer rank. No floating-point arithmetic occurs in the kernel.

The throughput benefit depends on batch size. A single query is slower on the GPU due to dispatch overhead. Ten thousand queries in parallel is where the GPU wins. In geometric applications—all vertices of a structure, all edge lengths, all angles—batches of this size arise naturally.

Mode 1 does not change what is being computed; it changes where and how fast.

### 8.2 Mode 2 — The $\varphi$-Native Crystal Computer

Mode 2 is different in kind, not just degree.

Each GPU thread is permanently assigned to one lattice node. The node's address is a six-vector $[a, b, c, d, e, f]$ with all non-negative integer components (the positive cone). The thread knows its six neighbors, reached by incrementing one component by one. This adjacency list is fixed at initialization and never changes.

Each thread maintains two state values: `last` and `next`. At each clock tick the thread:

1. reads `last` from all six neighbors,
2. computes its own `next` from those values, and

3. waits at a synchronization barrier.

When all threads have finished, the barrier releases, `next` becomes `last`, and the cycle repeats.

The CPU's only roles are loading initial conditions, starting the clock, and reading final state. Between clock ticks the CPU is idle. The lattice runs itself.

The Metal shader kernel is:

```
kernel void lattice_step(
    device PhiNode* last  [[ buffer(0) ]],
    device PhiNode* next  [[ buffer(1) ]],
    device int*     adj   [[ buffer(2) ]],
    uint   id             [[ thread_position_in_grid ]])
{
    int n0=adj[id*6+0], n1=adj[id*6+1], n2=adj[id*6+2],
        n3=adj[id*6+3], n4=adj[id*6+4], n5=adj[id*6+5];

    next[id].value = update(
        last[n0], last[n1], last[n2],
        last[n3], last[n4], last[n5]);
}
```

The adjacency buffer `adj` is precomputed from the six-basis arithmetic and uploaded once. The kernel itself contains only integer array reads, the `update` function, and a write.

## 8.3 Why Mode 2 Is a New Computational Model

A CPU cannot perform this computation, not because of speed but because the computation *is* the simultaneous local interaction of every node with its six neighbors. That has no sequential equivalent.

The update function is left open deliberately. It could encode:

- a wave equation (acoustic or electromagnetic propagation),
- a stress or strain field computation,
- a signal routing or switching rule, or
- a physics simulation native to quasicrystal geometry.

The lattice provides the geometry. The update function provides the physics. Together they define a programmable quasicrystal computer whose wiring diagram is the dodecahedral lattice itself.

A signal propagating through this lattice travels along $\varphi$-scaled paths and reflects off $\varphi$-symmetric boundaries. The propagation properties are those of physical quasicrystals—

directional, non-dispersive along the icosahedral axes—arising here not from material science but from the algebraic structure of the lattice.

### 8.4 Comparison of the Two Modes

| Property | Mode 1 | Mode 2 |
|---|---|---|
| Role of GPU | serves CPU | *is* the computation |
| Table used | Full $\mathbb{Z}[\varphi]$ | Positive cone only |
| Arithmetic | Signed | Non-negative |
| CPU during run | collects results | idle |
| Speedup question | yes (batch size) | not applicable |
| Novelty | engineering | new model |

Mode 1 is useful. Mode 2 is novel.

## 9 Summary and Future Work

We have presented four interlocking results.

**1. The digital slide rule.** PhiBase notation $P(p, n) = p\varphi + n$ with integer $p, n$ supports exact arithmetic via two precomputed tables. The Fibonacci Spine reduces multiplication to index addition; the Dense Grid reduces comparison to rank comparison. Any function with range in $[0, 1]$ is encoded at $10^{-8}$ precision. No floating-point arithmetic is required during computation.

**2. The integrality theorem.** Reflection through any of the six dodecahedral face-normal planes maps $\mathbb{Z}[\varphi]^3$ into $\frac{1}{5}\mathbb{Z}[\varphi]^3$, with denominator exactly 5 in all cases (Theorem 6.4). The six-basis inverse mapping absorbs this factor, restoring integer lattice coordinates. Reflections therefore reduce to $6 \times 6$ integer matrix multiplication at runtime.

**3. The $\varphi$-native GPU lattice.** The dodecahedral six-basis maps directly onto GPU thread space. Mode 1 provides batch arithmetic acceleration. Mode 2 creates a new computational model—a programmable quasicrystal computer—in which the lattice topology is the wiring diagram and each GPU thread is a permanently assigned lattice node.

Future work includes:

1. Explicit derivation of the six $6 \times 6$ reflection matrices and verification that they generate $I_h$.
2. Implementation of the Metal Mode 2 kernel and measurement of propagation properties for wave and stress update functions.
3. Extension to the full $\mathbb{Z}[\varphi]$ ring (signed coefficients) for Mode 1 table generation, with analysis of the required table depth for target precision.
4. Investigation of whether the $\frac{1}{5}\mathbb{Z}[\varphi]$ intermediate in reflection has physical significance in quasicrystal physics.

# A Arithmetic in $\mathbb{Z}[\varphi]$

Multiplication in $\mathbb{Z}[\varphi]$ uses $\varphi^2 = \varphi + 1$:

$$
\begin{aligned}
(p_1\varphi + n_1)(p_2\varphi + n_2) &= p_1p_2\varphi^2 + (p_1n_2 + n_1p_2)\varphi + n_1n_2 \\
&= p_1p_2(\varphi + 1) + (p_1n_2 + n_1p_2)\varphi + n_1n_2 \\
&= (p_1p_2 + p_1n_2 + n_1p_2)\,\varphi + (p_1p_2 + n_1n_2).
\end{aligned}
$$

Division uses the conjugate $\overline{p\varphi + n} = -p\varphi + (p + n)$:

$$
\frac{p_1\varphi + n_1}{p_2\varphi + n_2} = \frac{(p_1\varphi + n_1)\overline{(p_2\varphi + n_2)}}{(p_2\varphi + n_2)\overline{(p_2\varphi + n_2)}} = \frac{(p_1\varphi + n_1)(-p_2\varphi + p_2 + n_2)}{\mathcal{N}(p_2\varphi + n_2)},
$$

where $\mathcal{N}(p\varphi + n) = n^2 + np - p^2 \in \mathbb{Z}$. For $\mathbf{n} \cdot \mathbf{n} = P(1,\,2)$, we have $\mathcal{N} = 5$, establishing Lemma 6.2.

# B The Decode Vocabulary in Context

The seven symbols of Table 1 span coordinates $\{0, \pm1, \pm\varphi, \pm1/\varphi\}$. These are the only values that appear as vertex coordinates in any of the nine regular polyhedra (five Platonic solids and four Kepler–Poinsot solids) built from golden triangles. The vocabulary is complete for this geometry.